

Charotar University of Science and Technology

Education Campus, Changa – 388 421.

Biotech

M Sc (MICROBIOLOGY/BIOCHEMISTRY)

SEMESTER II EXAMINATION

BT 710: COMPUTATIONAL SKILLS

Date: 24.04.2010

Time: 10.00 AM to 11.30 AM

Total Marks: 35

Instructions to the candidates:

1. Attempt all questions
2. Questions in Part A should be answered in the question paper itself.
3. Write answers for Section I and Section II of Part B in separate answer sheets.
4. Assume suitable data wherever required.

PART A

Total marks: 10

Q1. Choose the correct option and put $\checkmark$ mark in front of it:

(1) What is the full form of ROM?

- | | |
|-----------------------|-------------------------|
| a. Random Only Memory | c. Random Access Memory |
| b. Read Only Memory | d. None of the above |

(2) What is the extension of database file in Microsoft Excel?

- | | |
|--------|--------|
| a. mdf | c. xcl |
| b. xls | d. xel |

(3) Which C function will print the message on the display?

- | | |
|--------------|-------------------|
| a. printf(); | c. printscreen(); |
| b. scanf(); | d. printmsg(); |

(4) Which of the following is Application software?

- | | |
|---------------------|---------------------|
| a. Microsoft Office | c. Linux |
| b. Windows Vista | d. Turbo C Compiler |

(5) Which of the following is full name of WWW?

- | | |
|-------------------|-----------------------|
| a. World wide web | c. World website wide |
| b. World web wide | d. World wise web |

(6) Which of the following is not a search engine website?

- | | |
|-------------------|-------------------|
| a. www.google.com | c. www.bing.com |
| b. www.yahoo.com | d. www.amazon.com |

(7) _____ from following is not an input device?

- | | |
|-------------|----------------|
| a. Keyboard | c. Joy stick |
| b. Plotter | d. Digital pen |

(8) If $f(x) = x^2 + 2x + 1$, then $f(x-1) =$ _____.

(9) If $\cos A = 3/5$, then $\sin A =$ _____.

(10) Which C function is used to get the input from user at the time of execution of Program?

- | | | | |
|----------|-----------|----------|-----------|
| a. scanf | b. printf | c. getch | d. clrscr |
|----------|-----------|----------|-----------|

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Part B

Total marks: 25

Section I

Q1. Attempt any two:

(5)

(i) If $f(x) = \log x$, show that

(a) $f(mn) = f(m) + f(n)$

(b) $f(m/n) = f(m) - f(n)$

(c) $f(x^n) = n \log x$

(ii) Prove that $\tan 9^\circ + \tan 36^\circ + \tan 9^\circ \tan 36^\circ = 1$

(iii) If $A = \begin{pmatrix} 0 & 2 & 3 \\ 2 & 1 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 7 & 6 & 5 \\ 1 & 4 & 5 \end{pmatrix}$

Find the matrices (a) $2A + 3B$, (b) $3A - B$, (c) AB

Q-2 Attempt any one:

(5)

(i) Explain in brief about RAM and write a short note on "Mainframe Computer."

(ii) Draw the general architecture of Computing System.

Section II

Q1. Attempt any one:

(5)

(i) Explain in brief about DBMS. How it is differ from traditional file system?

(ii) Draw the architecture of database system in context of view of data.

Q2. Attempt any one:

(5)

(i) Write a short note on preprocessor and preprocessor directives in 'C'.

(ii) Write a program for addition, subtraction, multiplication and division of two numbers in single C program.

Q3. Attempt any one:

(5)

(i) Write a short note on: Data warehouse and Data Mining.

(ii) What is Bioinformatics? List out activities in Bioinformatics and explain it in brief.